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(54) VACUUM PACKAGE FOR FOOD PRODUCTS

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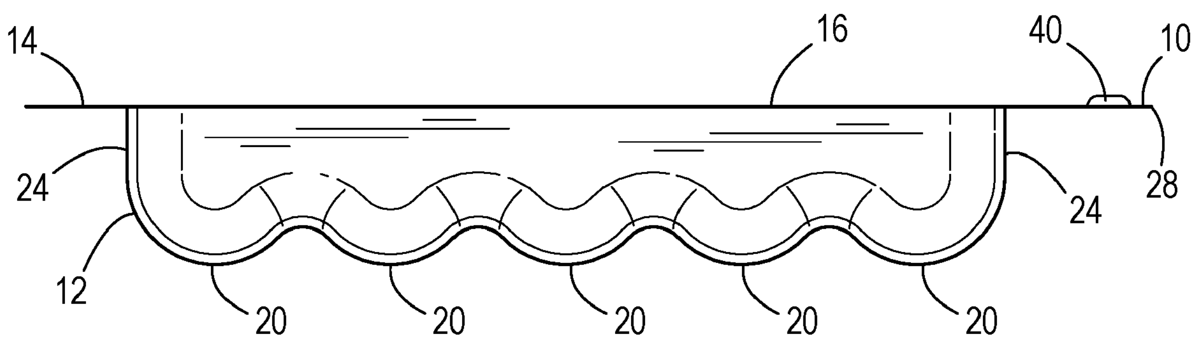
(57) ABSTRACT

A vacuum package for containing a plurality of food products. The package includes a compartment including a plurality of pockets defined by semi-cylindrical walls. One food product is housed in each pocket. A cover encloses the compartment to define an interior. A vacuum is applied to the interior. The shape of the compartment enables the package to be more easily filled using a horizontal vacuum pack machine on a manufacturing line.



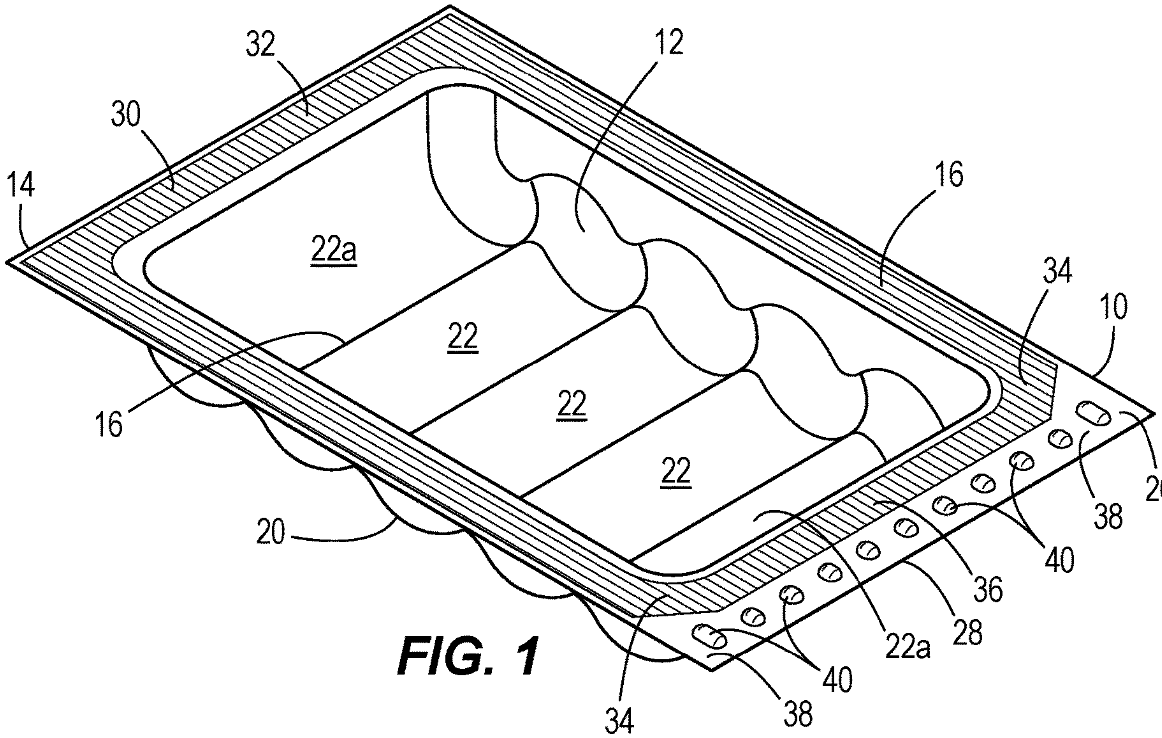

32
12
30
14
22a
16
34
10
22
16
22
22
26
38
20
36
40
22a
28
40
38
34

FIG. 1

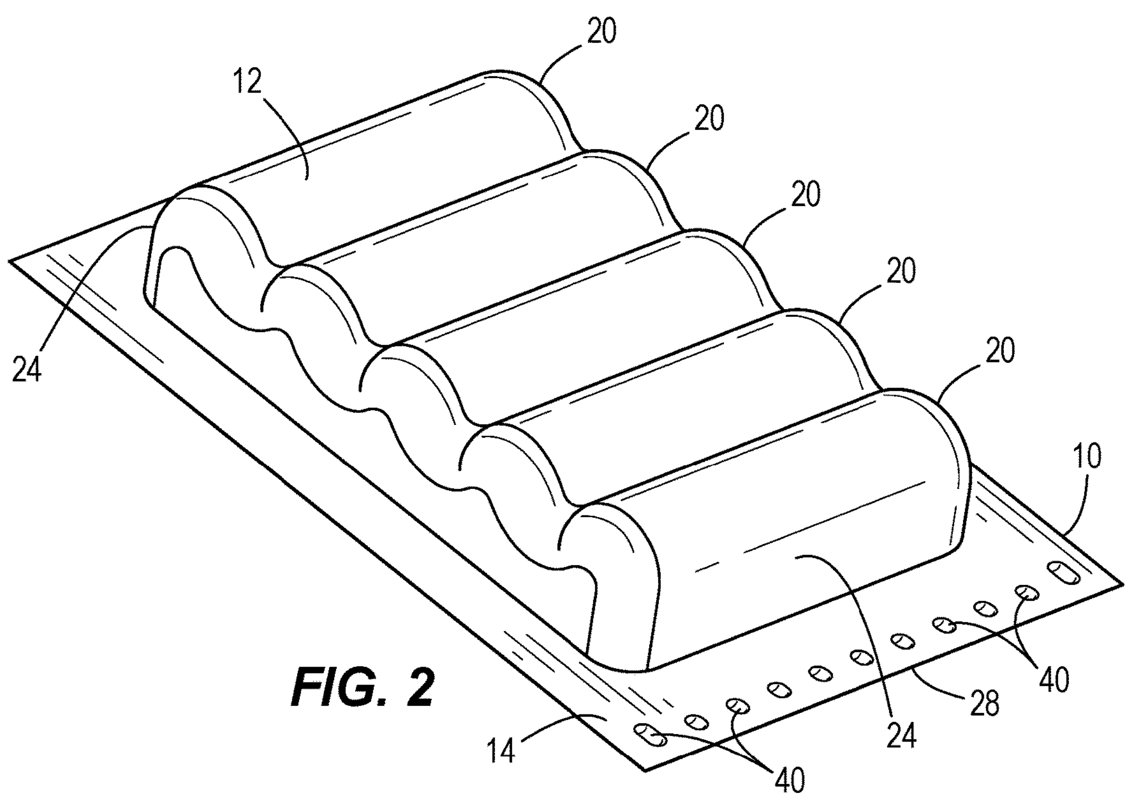

20
12
20
20
20
20
24
10
40
28
24
14
40

FIG. 2

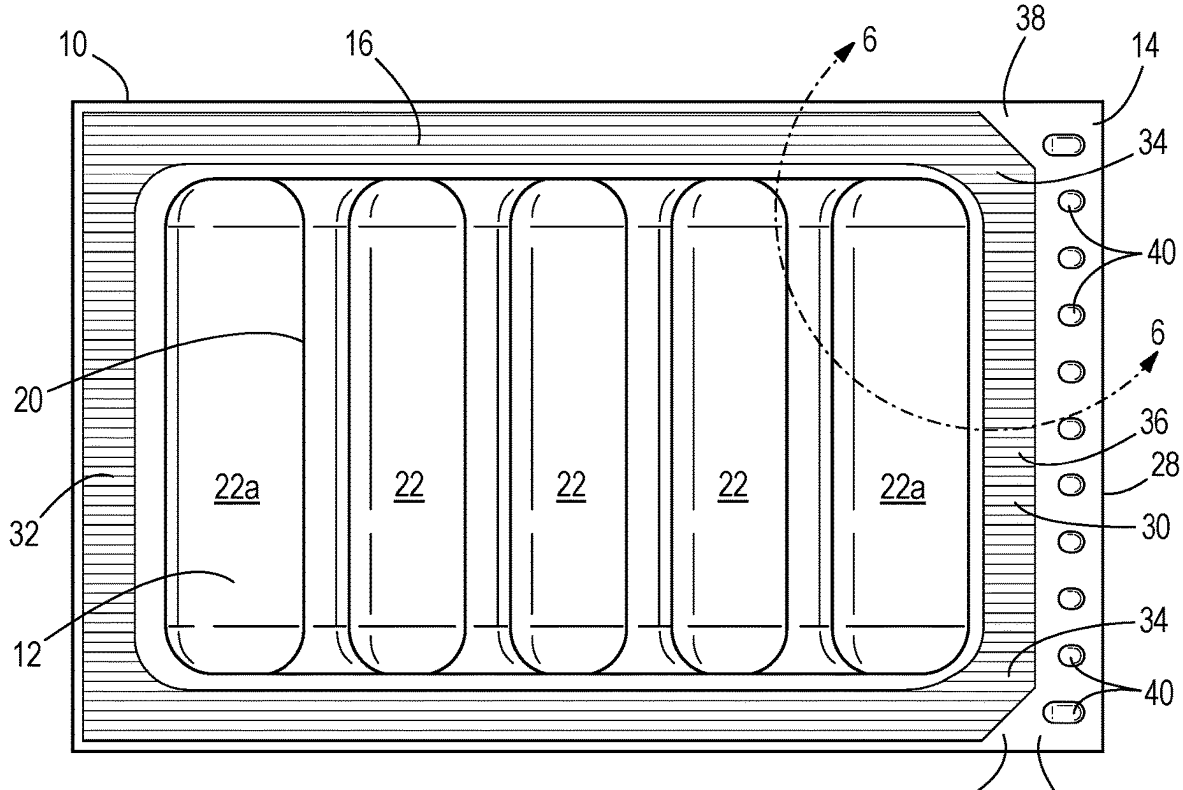

10
16
6
38
14
34
40
20
6
36
28
22a
22
22
22
22a
32
30
12
34
40
26
38

FIG. 3

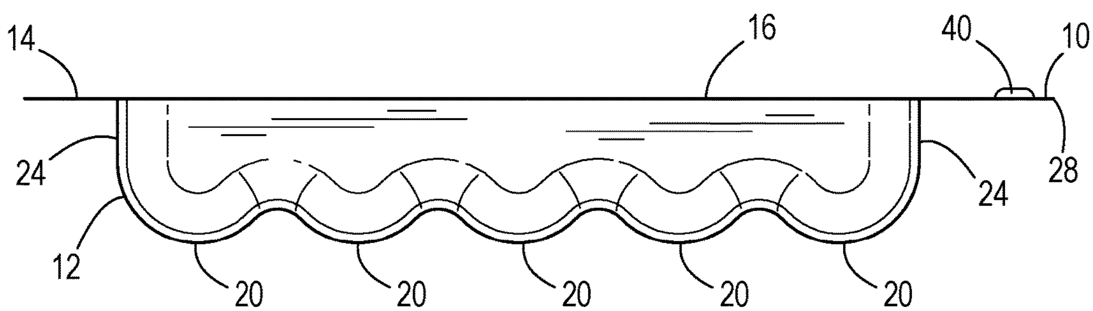

14
16
40
10
24
28
24
12
20
20
20
20
20

FIG. 4

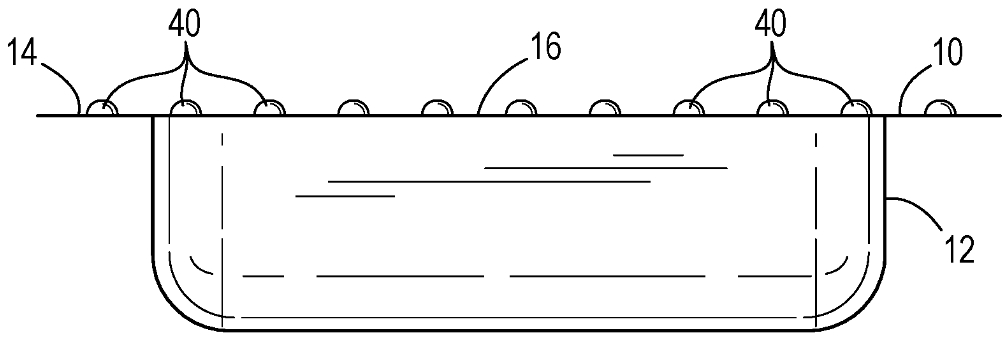


FIG. 5

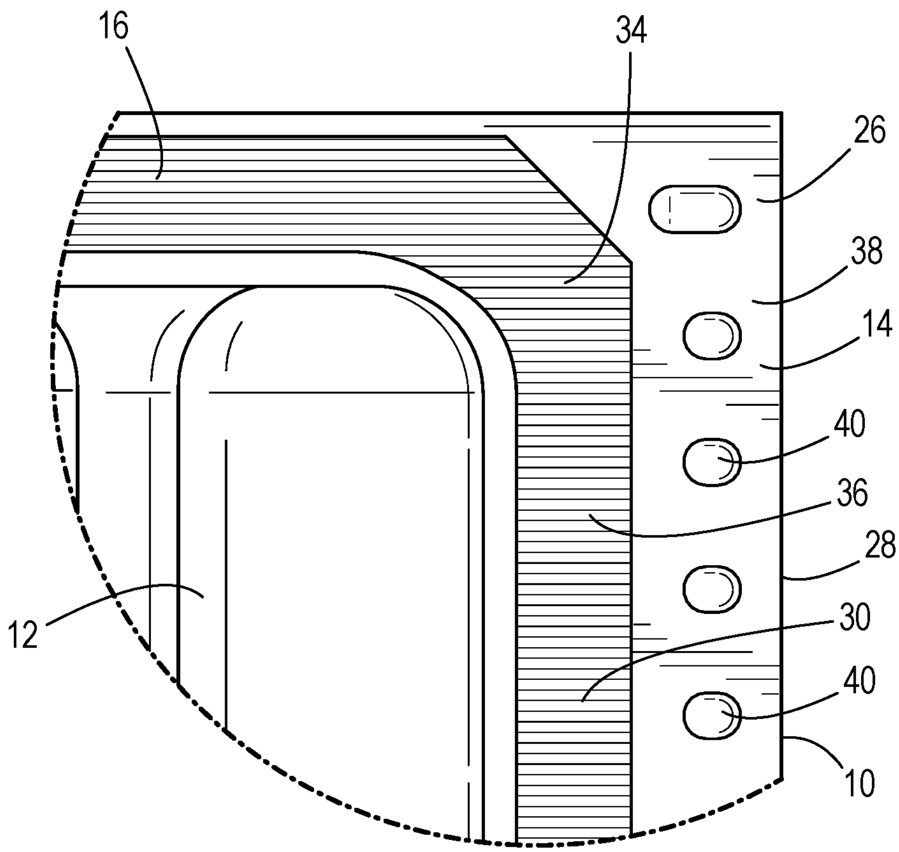


FIG. 6

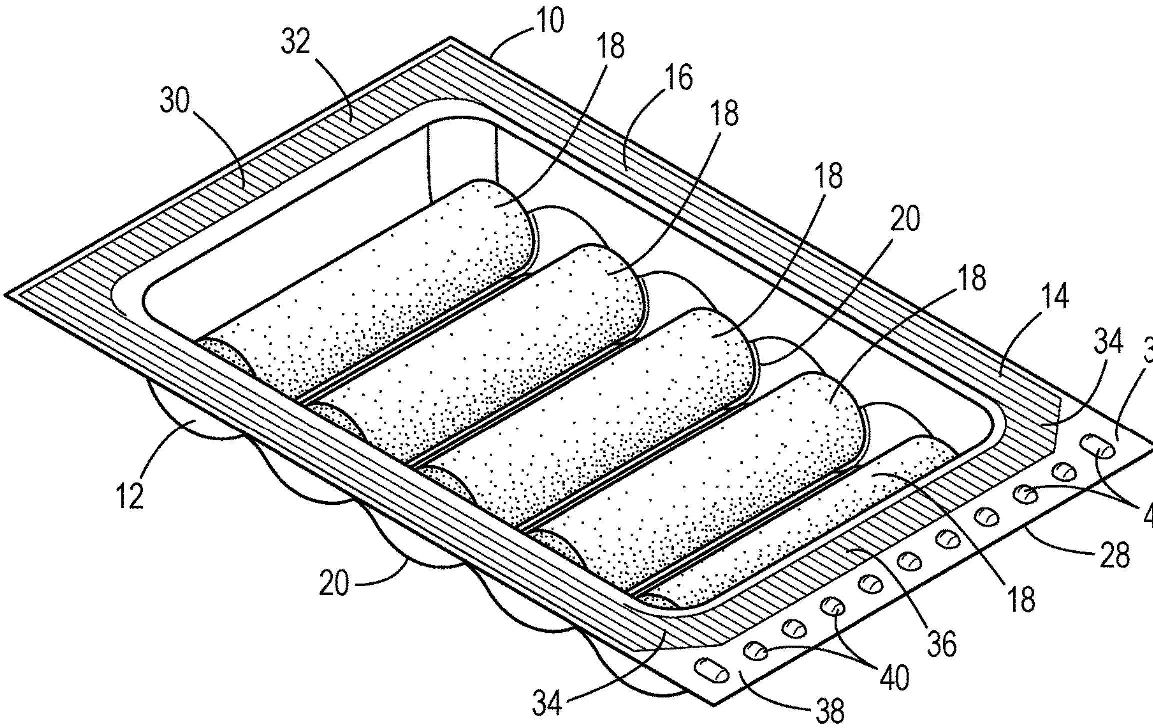


FIG. 7

VACUUM PACKAGE FOR FOOD PRODUCTS

FIELD OF THE INVENTION

[0001] The present disclosure is in the field vacuum packages, and more specifically, vacuum packages containing food products that are sealable on vacuum pack machines to allow for the freshest possible food products.

BACKGROUND OF THE INVENTION

[0002] Vacuum packages are commonly utilized in a manufacturing line and are filled with food products using a horizontal vacuum pack machine. Current vacuum packages have the disadvantage of needing extra labor to align and/or to set the food products in the vacuum package so that they are appropriately seated and/or aligned in the vacuum package. Another disadvantage relates to the amount of air surrounding the food products in vacuum packages. With room for air surrounding the food products, seals may not be as secure and shelf life reduced.

SUMMARY OF THE INVENTION

[0003] This invention relates to a vacuum package for containing a plurality of food products comprising a compartment including a plurality of pockets defined by semi-cylindrical walls, a plurality of food products with one food product housed in each pocket, a cover enclosing the compartment to define an interior, and a vacuum in the interior.

[0004] This invention relates to a vacuum package for containing a plurality of food products comprising a compartment having a plurality of elongate parallel pockets defined by semi-cylindrical walls and having a planar top surface, a plurality of food products with one food product housed in each pocket, a cover heat sealed to the top surface and enclosing the compartment to define an interior, and a vacuum in the interior.

[0005] The invention relates to a vacuum package for housing a plurality of cylindrical food products comprising a compartment including a plurality of elongate parallel pockets defined by semi-cylindrical walls, the compartment including a top surface and a periphery, a plurality of cylindrical food products with only one food product housed in each pocket, a flexible film heat sealed to the top surface to enclose the compartment and define an interior, the flexible film has a periphery that is the same shape as the periphery of the compartment, and having a portion that is non-heat sealed at the periphery of the compartment so as to be lifted from the periphery of the compartment to gain access to the interior, and a vacuum in the interior.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a perspective view of a top of a vacuum package of the present invention.

[0007] FIG. **2** is a perspective view of a bottom of the vacuum package.

[0008] FIG. **3** is a top view of the vacuum package.

[0009] FIG. **4** is a side view of the vacuum package.

[0010] FIG. **5** is an end view of the vacuum package

[0011] FIG. **6** is a detailed corner view of FIG. **3**.

[0012] FIG. **7** is a perspective view of the top of the vacuum package containing food products.

[0013] Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of constructions and the arrangement of components set forth in the following description or illustrated in the drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways.

DETAILED DESCRIPTION OF THE INVENTION

[0014] FIGS. **1**-**7** illustrate a vacuum package **10** according to the present invention. The vacuum package **10** includes a specifically shaped compartment **12**, a top wall **14** and a cover **16**. The overall shape of the vacuum package **10** and the specific shape of the compartment **12** enable the vacuum package **10** to be more easily filled with a food product and then sealed on a horizontal vacuum pack machine on a manufacturing line.

[0015] The vacuum package **10** is designed to be filled with a food product, such as a perishable food product. The invention will be further detailed below using a cylindrical perishable food product **18** (FIG. **7**) such as for example string cheese sticks. However, it should be noted that food products and other perishable food products of different shapes that need vacuum sealing can also be used.

[0016] The compartment **12** includes a plurality of generally semi-cylindrical walls or ribs **20**. Each rib **20** defines a pocket **22**. All of the pockets **22** are elongate, parallel to one another, and semi-cylindrical in shape. The pockets **22*a*** on the ends of the compartment **12** further include an end wall **24** to close off the compartment **12**. As shown, the compartment **12** includes two end pockets **22*a*** and three interior pockets **22**, but other numbers of pockets and other orientations can also be utilized. Each rib **20** enables one food product **18** to be placed in one pocket **22** and enables the food product **18** to settle into that pocket **22** during manufacture allowing for more flexibility in robotic loading in that each food product **18** has a designated pocket **22** and each food product **18** is unable to settle at an angle in the pocket **22**, thus eliminating the need for manual labor to load the food products **18** into the compartment **12**

[0017] The ribs **20** can have a radius of curvature of generally R7.8 mm with the upper ridge R3, however, other radii can be used as needed. The pockets **22** are particularly suited for a cylindrical food product **18**, however, it should be noted that other shaped perishable and non-perishable food products can also be utilized to be contained in the pockets **22**, such as rectangularly shaped food products.

[0018] The top wall **14** is preferable planar to provide a surface **26** onto which to secure the cover **16**. The top wall **14** is preferably has a rectangular periphery with an end **28** but other shapes can also be utilized.

[0019] The compartment **12** and the top surface **26** are preferable flexible and manufactured from a material, such as for example, multilayer barrier film with a polyester outer and barrier layer, however, other materials can also be utilized.

[0020] As best shown in FIGS. **1**, **3**, **6** and **7**, the cover **16** encloses the compartment **12**. Preferably, the cover **16** has a periphery that matches the periphery of the top surface **26**, which is shown as rectangular but other shapes can also be utilized. Preferably, the cover **16** is a flexible film such as for example, multilayer barrier film with a polyester outer and barrier layer, however, other types of flexible film can also be utilized. The cover **16** is sealed to the top surface such as preferably by heat sealing. An example of a pattern **30** of sealing the cover **16** to the top surface **26** is shown and is generally rectangular on one end **32** and having curved portions **34** at the other end **36**. As best shown in FIGS. **3**, **6** and **7**, the cover **16** is sealed to the top surface **26** in the pattern **30** to create two generally triangular portions **38** of the cover **16** that act as peeling corners. A consumer is able to access the food products **18** in the compartment **12** by lifting up on one or both triangular portions **38** of the cover **16**, by lifting up on the cover **16** adjacent the end **28** of the top wall **14** or by lifting up on any portion of the cover **16** to break the seal of the cover **16** and gain access to the food products **18**.

[0021] A vacuum is applied to the interior of vacuum package **10** as is known in the art during manufacture.

[0022] With the present invention, the ribs **20** affirmatively align and allow setting of the food products **18** in the pockets **22** of the compartment **12** eliminating the need for extra labor and extra costs to seat the food products. The ribs **20** create a form for the food products **18** to be housed in that has less room for air in the compartment **12**. The less room for air equates to better seals and potentially better shelf life.

[0023] To assist the consumer with opening the vacuum package **10**, peel bumps **40** can optionally be utilized. A plurality of peel bumps **40** are preferably aligned in a row on the top wall **14** adjacent the end **28**. As shown, there are nine semi-spherical and two semi-oblong peel bumps **40**, however, it should be noted that other numbers and shapes of peel bumps **40** can be utilized. The peel bumps **40** assist the consumer in grasping the cover **16** to aid the consumer in opening the vacuum package **10** to access its food products **18**.

[0024] Various features and advantages of the invention are set forth in the following claims.

1. A vacuum package for containing a plurality of food products comprising:

a compartment including a plurality of pockets defined by semi-cylindrical walls;

a plurality of food products with one food product housed in each pocket;

a cover enclosing the compartment to define an interior; and

a vacuum in the interior.

2. The vacuum package of claim **1** and further including a top surface capable of securing the cover thereto.

3. The vacuum package of claim **2** wherein the top surface is planar.

4. The vacuum package of claim **1** wherein the cover includes a flexible film.

5. The vacuum package of claim **1** wherein the food products are of a complementary shape to the pockets.

6. The vacuum package of claim **1** wherein the food products are cylindrically shaped.

7. The vacuum package of claim **1** wherein the cover is heat sealed to the compartment.

8. The vacuum package of claim **2** wherein the cover is heat sealed to the top surface.

9. The vacuum package of claim **8** wherein the compartment further includes a top surface and wherein the cover is heat sealed to the top surface.

10. The vacuum package of claim **1** wherein the compartment has a periphery and wherein the cover has the same periphery.

11. The vacuum package of claim **10** wherein the cover is heat sealed to the compartment with at least one portion of the heat seal being inward of the periphery of the compartment.

12. The vacuum package of claim **10** wherein the cover has at least one portion that is not heat sealed to the compartment and is useable to peel the cover from the compartment.

13. The vacuum package of claim **1** wherein the compartment has a periphery that is rectangular.

14. The vacuum package of claim **1** wherein the cover has a periphery that is rectangular.

15. The vacuum package of claim **1** wherein the compartment and the cover have peripheries that are the same shape.

16. The vacuum package of claim **1** wherein the pockets are elongate and parallel to one another.

17. A vacuum package for containing a plurality of food products comprising:

a compartment having a plurality of elongate parallel pockets defined by semi-cylindrical walls and having a planar top surface;

a plurality of food products with one food product housed in each pocket;

a cover heat sealed to the top surface and enclosing the compartment to define an interior; and

a vacuum in the interior.

18. The vacuum package of claim **17** wherein the food products are elongate and cylindrical.

19. The vacuum package of claim **17** wherein the cover includes a flexible film.

20. A vacuum package for housing a plurality of cylindrical food products comprising:

a compartment including a plurality of elongate parallel pockets defined by semi-cylindrical walls, the compartment including a top surface and a periphery;

a plurality of cylindrical food products with only one food product housed in each pocket;

a flexible film heat sealed to the top surface to enclose the compartment and define an interior, the flexible film has a periphery that is the same shape as the periphery of the compartment, and having a portion that is non-heat sealed at the periphery of the compartment so as to be lifted from the periphery of the compartment to gain access to the interior; and

a vacuum in the interior.

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